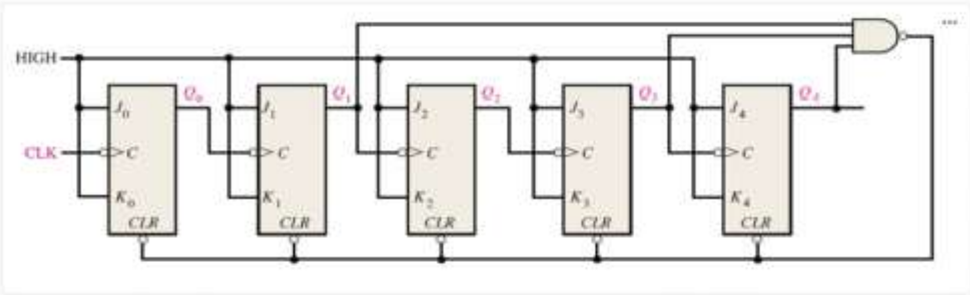
JK FLIPFOP FREQUENCY ANSWR

The frequency of the clock waveform input to the circuit shown below is 10 MHz. determine the frequency, in kHz, of the output waveform at Q_4. Round your answer to the nearest whole number.



Round your answer to 0 decimal places.

Your answer: 4

Incorrect

The answer is **385**

<u>Theory</u>

- In **Synchronous operation**: All flip-flops receive the **same clock**, eliminating ripple-delay (all outputs change together) therefore the flip flops are synchronised

- **Truncated sequence**

Asynchronous Decade Counters

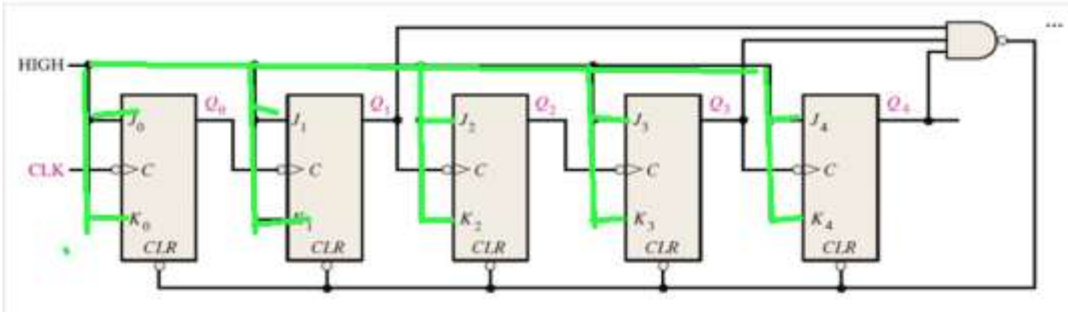
The **modulus** of a counter is the number of unique states through which the counter will sequence. The maximum possible number of states (maximum modulus) of a counter is 2^n, where n is the number of flip-flops in the counter. Counters can be designed to have a number of states in their sequence that is less than the maximum of 2^n. This type of sequence is called a *truncated sequence*.

  - They also know as Modulo counters (MOD-N): These counters out up to a predefined number N and then reset to Zero
  - They do not run the full 2^n states
- So they can run the full states but sometimes we change that as in make the restart earlier

- As we observe the J-K flipflop we can see a few things which is the
  - All JKs are tied J=K=1 → they toggle on every clock edge.
    -

Truth Table

J	K	CLK	Q
0	0	↑	Q_0 (no change)
1	0	↑	1
0	1	↑	0
1	1	↑	$\overline{Q}_0$ (toggles)

- 

- The **CLR** pin has a little **bubble (small circle)** at its input.
  - That bubble means the input is **active-LOW**.

  - In simple terms to activate the clear we have to input a low which is a zero
  - This will clear the flipflops (Restart them)

- Now look at the and gate

3-Input NAND Gate

Truth Table

Input A	Input B	Input C	X = (A.B.C)'
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

  - It only gives us a 0 when all inputs are high
  - The output of the NAND gate is connected to clear of all the flipflops
  - So at the point where the NAND gate receives all 1s then it will restart the count

- The inputs to the gate are Q1, Q3, Q4 when all of them are one thats when the count will restart
- To get the limit in decimal we set the all Q pins to 0 except the one connected to the NAND gate do from Q4-Q0 (MSB-LSB)
- 11010 which is 26 in decimal that is the limit

- The counter counts from `00000` (0) up to `11001` (25)
- At `11010` (26), the AND gate activates → drives CLR LOW → all FFs reset to 0.
- This makes the counter repeat every 26 cycles.

Frequency division

Further division of a clock frequency can be achieved by using the output of one flip-flop as the clock input to a second flip-flop, as shown in Figure 7–37. The frequency of the Q_A output is divided by 2 by flip-flop B. The Q_B output is, therefore, one-fourth the frequency of the original clock input. Propagation delay times are not shown on the timing diagrams.

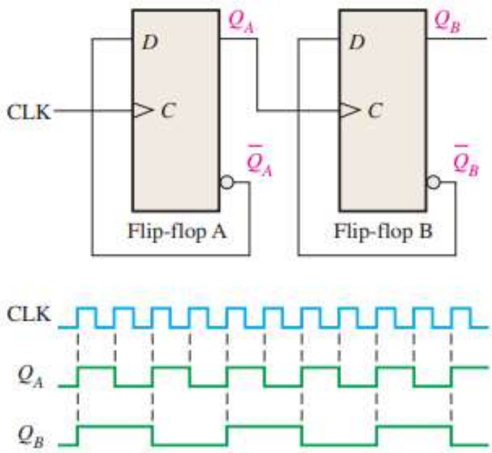


FIGURE 7–37 Example of two D flip-flops used to divide the clock frequency by 4. Q_A is one-half and Q_B is one-fourth the frequency of CLK. Open file F07-37 and verify the operation.

MultiSim

By connecting flip-flops in this way, a frequency division of 2^n is achieved, where n is the number of flip-flops. For example, three flip-flops divide the clock frequency by $2^3 = 8$; four flip-flops divide the clock frequency by $2^4 = 16$; and so on.

EXAMPLE 7–9

Develop the f_{out} waveform for the circuit in Figure 7–38 when an 8 kHz square wave input is applied to the clock input of flip-flop A.

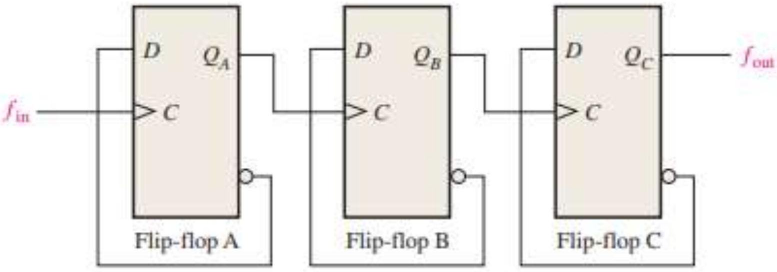


FIGURE 7–38

Solution

The three flip-flops are connected to divide the input frequency by eight ($2^3 = 8$) and the Q_C (f_{out}) waveform is shown in Figure 7–39. Since these are positive edge-triggered flip-flops, the outputs change on the positive-going clock edge. There is one output pulse for every eight input pulses, so the output frequency is 1 kHz. Waveforms of Q_A and Q_B are also shown.

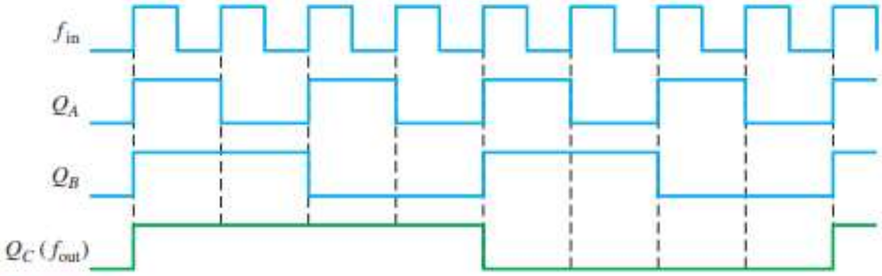


FIGURE 7–39

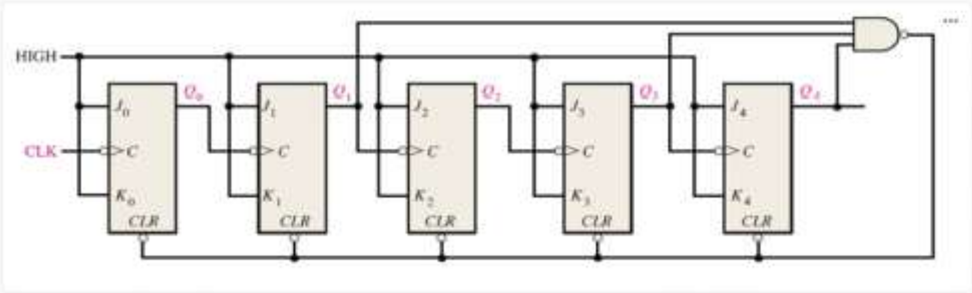
Related Problem

How many flip-flops are required to divide a frequency by thirty-two?

Therefore

$$output\ frequency = \frac{input\ clock}{modulus}$$

The frequency of the clock waveform input to the circuit shown below is 10 MHz. determine the frequency, in kHz, of the output waveform at Q_4. Round your answer to the nearest whole number.



Round your answer to 0 decimal places.

Your answer: 4

Incorrect

The answer is **385**

ANSWER

$$outputfrequency = \frac{input\ clock}{modulus} = \frac{10\text{Mhz}}{26} = 384615.3846 = 384.615x10^3 = 385Khz$$

19:14 72

Test

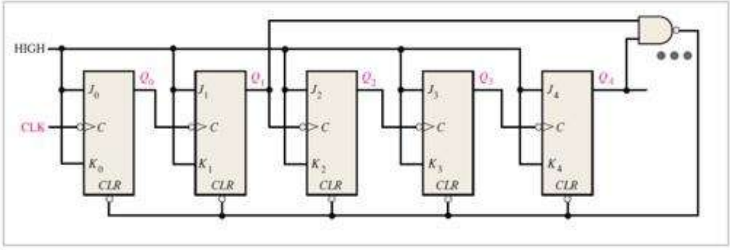
Correct

The answer is **3**

8 **CALCULATED FORMULA** 0 / 7

INCORRECT

The frequency of the clock waveform input to the circuit shown below is 11 MHz. determine the frequency, in kHz, of the output waveform at Q_4. Round your answer to the nearest whole number.



Round your answer to 0 decimal places.

Your answer: 344

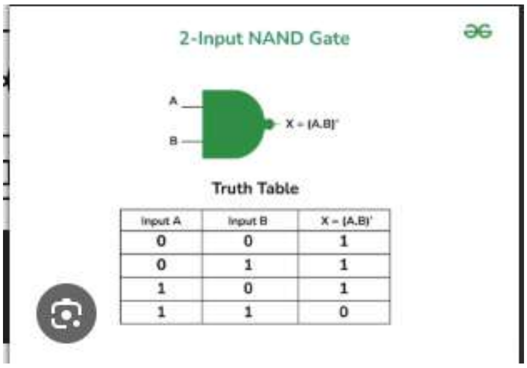
Incorrect

The answer is **611**

9 **CALCULATED NUMERIC** 0 / 10

INCORRECT

- Two input nand gate



-
- Q1& Q4 must be 1 to clear
- MSB-LSB 10010
- DEC = 18

ANSWER

$$outputfrequency = \frac{input\ clock}{modulus} = \frac{11Mhz}{18} = 611111.111 = 611.11110^{3} = 611Khz$$